

Customer Churn Data Analytics Project

Stage 2 Summary: Data Acquisition & Documentation

This document summarizes the Data Acquisition & Documentation stage of the Customer Churn Data Analytics Project. This stage is documentation-focused rather than analytical: no churn rates are calculated and no relationships between variables are investigated here. The purpose is to establish exactly what data is available, where it came from, what each column represents, and how the dataset will be used going forward.

1. Stage Activities

Step	Task	Description	Tool
2.1	Identify Data Source	Record the original source of the dataset.	Word / Google Docs
2.2	Obtain the Dataset	Load/download the raw customer churn dataset and preserve the original version unchanged.	Python, Pandas
2.3	Record Dataset Structure	Document number of rows, number of columns, and file format.	Python, Pandas
2.4	Document All Variables	Create a data dictionary describing every column, its meaning, and its expected type.	Excel / Google Sheets
2.5	Identify Target Variable	Document churn as the primary target/business outcome.	Excel / Word
2.6	Classify Variables	Identify categorical, numerical, binary/flag, identifier, and target variables.	Excel / Python
2.7	Identify Data Granularity	Establish what one row represents (in this case, one customer).	Word / Excel
2.8	Preserve Raw Data	Keep an untouched copy of the original dataset for reproducibility and comparison.	File system / Excel
2.9	Document Initial Assumptions	Record assumptions about variable meanings, encoding, units, and business interpretation before analysis.	Word / Excel
2.10	Document Dataset Limitations	Record known limitations such as available variables, missing business information, observational nature of the data, etc.	Word

2. Data Dictionary

The table below documents every column in the dataset, its expected data type, category, and business meaning. This is one of the most important outputs of this stage. For the final version, only the columns actually present in the dataset should be included — variables should not be added simply because they exist in a common version of the Telco Churn dataset.

Column	Data Type	Category	Business Meaning
customerid	String	Identifier	Unique customer identifier
gender	Categorical	Demographic	Customer gender
seniorcitizen	Integer	Binary	Whether the customer is a senior citizen
partner	Categorical	Demographic	Whether the customer has a partner
dependents	Categorical	Demographic	Whether the customer has dependents
tenure	Integer	Numerical	Number of months the customer has stayed
phoneservice	Categorical	Service	Whether the customer has phone service
multiplelines	Categorical	Service	Whether the customer has multiple lines
internetservice	Categorical	Service	Type of internet service
onlinesecurity	Categorical	Service	Whether online security is subscribed
onlinebackup	Categorical	Service	Whether online backup is subscribed
deviceprotection	Categorical	Service	Whether device protection is subscribed
techsupport	Categorical	Service	Whether technical support is subscribed
streamingtv	Categorical	Service	Whether streaming TV is subscribed
streamingmovies	Categorical	Service	Whether streaming movies is subscribed
contract	Categorical	Contract	Customer contract type
paperlessbilling	Categorical	Billing	Whether paperless billing is enabled
paymentmethod	Categorical	Billing	Customer payment method
monthlycharges	Numeric	Financial	Monthly amount charged
totalcharges	Numeric	Financial	Total amount charged
churn	Categorical	Target	Whether the customer churned

3. Variable Classification

Identifiers

- customerid

Numerical

- tenure
- monthlycharges
- totalcharges

Categorical

- gender
- partner

- dependents
- phoneservice
- multiplelines
- internetservice
- onlinesecurity
- onlinebackup
- deviceprotection
- techsupport
- streamingtv
- streamingmovies
- contract
- paperlessbilling
- paymentmethod

Binary / Flag

- seniorcitizen

Target

- churn

4. Data Granularity

Each row in the dataset represents a single customer. This unit of observation applies consistently across all variables and is the basis for all subsequent analysis.

5. Raw Data Preservation

An untouched copy of the original dataset is preserved separately from any working copy used for analysis, ensuring reproducibility and enabling comparison against the original source at any later stage.